

Screening the OPERA archive for km-scale precipitation extremes

Method, validation against gauges, and questions for MeteoSwiss

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Why the data matter for this project

Task. Super-resolution of precipitation from 25 km to 2 km over Europe (128×128 tiles of 256 km), deterministic and flow-matching models, with a structural loss built on Minkowski functionals of the rain field.

Evaluation targets the tail: exceedance ratios and return levels from a peaks-over-threshold fit at a fixed level (currently 31 mm h^{-1}), and scores on 27 documented storms.

So the data must provide:

- ▶ a tail made of rain, not radar artefacts: these statistics rest on a few thousand pixels, and one undetected radar fault was 12% of all test pixels above 89 mm h^{-1} ;
- ▶ splits that are independent in time, so test storms are not neighbours of training frames;
- ▶ test sets for shifts: documented extremes, and a change of radar product.

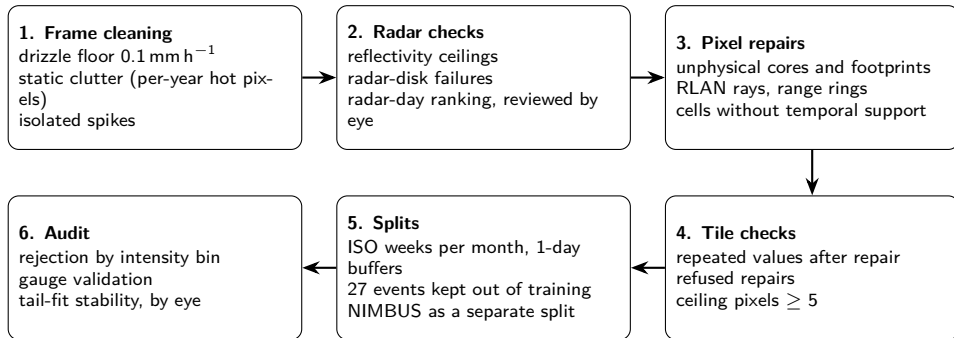
Data

- ▶ EUMETNET OPERA rain-rate composite with quality index, open archive: **5,023 days, 2012-09 to 2026**, LAEA grid 2200×1900 , 2 km, 15 min.
- ▶ **Two production chains:** ODYSSEY until 2024-07-05 (composite of the scans in a 15-min window), NIMBUS after (lowest-elevation scan at the nominal time). Earlier breaks: beam-blockage correction (2015), log $\rightarrow$ linear compositing (2017-09-29).
- ▶ **The raw field is dirty.** On a sample day (2018-06-15), 52 of 96 composites have a maximum above 500 mm h^{-1} ; the peak is $13,072 \text{ mm h}^{-1}$.
- ▶ Independent truth for validation: 10-min gauges from DWD (1,454) and SwissMetNet (281), 12.1M station-frames.

Principles of the screen

1. **Remove clear errors only; keep imperfections.** Blockage, attenuation, compositing seams and mild clutter stay, so the model learns to be robust to them.
2. **Key every rule on an error signature:** geometry (rays, rings, range-only fields), persistence, exact repeated values, physical impossibility. Never on intensity or “unusual structure” alone, because real extremes are unusual too.
3. **Repair pixels rather than drop tiles** where the artefact is local.
4. **No labelled set.** Instead: rejection rate per intensity bin for every rule, the strongest flags of each rule looked at by eye, and validation against gauges.
5. **The same screen for training and evaluation.**
6. **The tail fit is an audit, never a rule.** Removing whatever makes the GPD fit better would select the data by the model used to evaluate it.

Pipeline



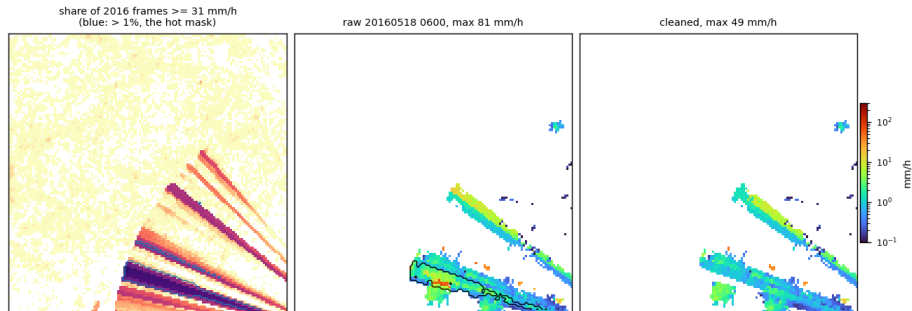
Radar checks run on the field before any repair, so a repair cannot hide a failure. All decisions are stored as per-tile columns and applied at split time, so a threshold can change without reprocessing the archive.

Frame cleaning and pixel repairs

rule	signature	action
static clutter	pixel $\geq 31 \text{ mm h}^{-1}$ in $> 1\%$ of a calendar year's frames	median of non-hot 5×5 neighbours
isolated spike	pixel $\geq 10 \text{ mm h}^{-1}$, every neighbour $< 10\%$ of it	neighbour maximum
unphysical core	pixel $> 500 \text{ mm h}^{-1}$, with its connected region $\geq 150 \text{ mm h}^{-1}$	median of the outer ring
RLAN ray	thin component $\geq 80 \text{ km}$ aligned ($< 10^\circ$) with a radar within 250 km	median of the outer ring
range ring	circle in the per-year $\geq 31 \text{ mm h}^{-1}$ frequency around a site; pixels $\geq 89 \text{ mm h}^{-1}$	median of the outer ring
no temporal support	cell $\geq 10 \text{ mm h}^{-1}$ with no echo $\geq 1 \text{ mm h}^{-1}$ within 30 km at $t \pm 15 \text{ min}$	median of the outer ring

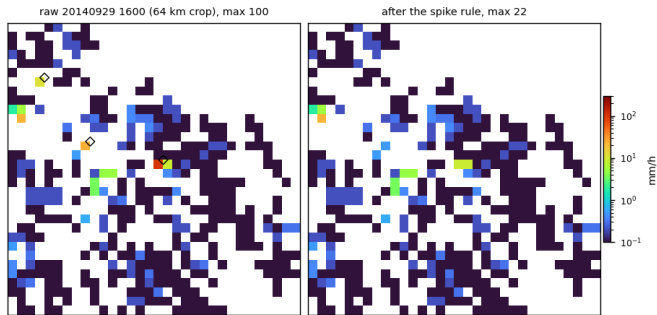
Nothing is zeroed. Tiles must be fully covered. The repair is guarded: a component above 500 pixels is not repaired, and the tile is rejected instead.

Example: static clutter



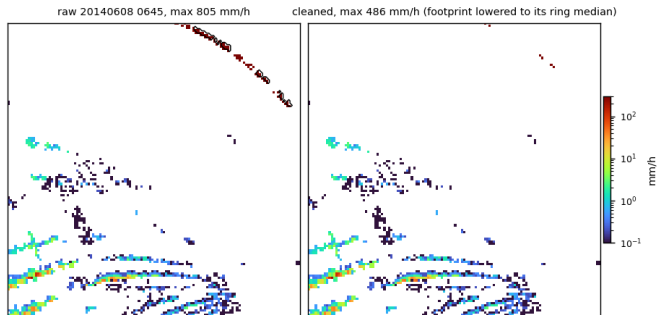
Fails: the same pixels exceed 31 mm h^{-1} in more than 1% of a year's frames; here the radial sectors of one radar in 2016 (left, blue: the hot mask). **Rule:** every rainy pixel of the mask is lowered to the median of the non-hot pixels in its 5×5 window, if that is lower. The rain band crossing the sector keeps its shape ($\text{max } 81 \rightarrow 49 \text{ mm h}^{-1}$).

Example: isolated spike



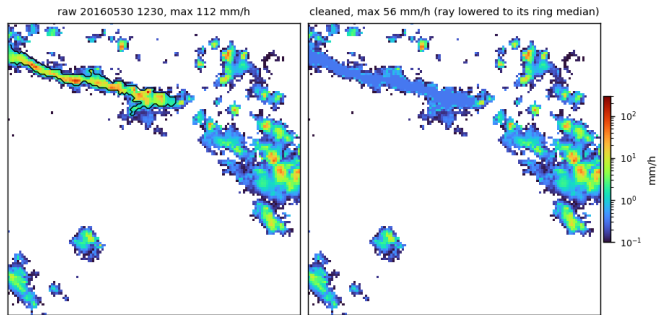
Fails: single pixels far above everything around them (diamonds, 64 km crop). **Rule:** a pixel $\geq 10 \text{ mm h}^{-1}$ whose 8 neighbours are all below 10% of it takes the largest neighbour's value: $100 \rightarrow 22 \text{ mm h}^{-1}$. **Limit:** two adjacent bad pixels protect each other, so small clusters survive (the Oradea case, later).

Example: unphysical values and their footprint



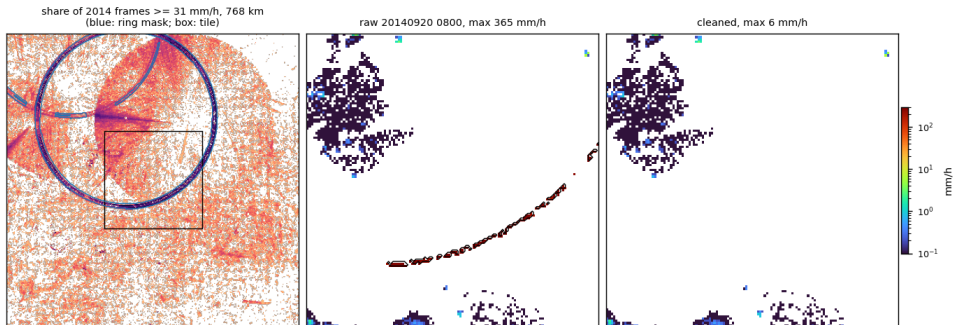
Fails: values above 500 mm h^{-1} on an arc at constant range from a radar (black). **Rule:** each pixel $> 500 \text{ mm h}^{-1}$, with the region $\geq 150 \text{ mm h}^{-1}$ connected to it, is lowered to the median of the region's outer ring; the rain below is untouched. **Left over:** 486.2, 452.5, 421.1, 391.8 mm h^{-1} on the same arc, i.e. 66.0, 65.5, 65.0, 64.5 dBZ: steps of 0.5 dBZ, not connected to a pixel above 500.

Example: RLAN ray



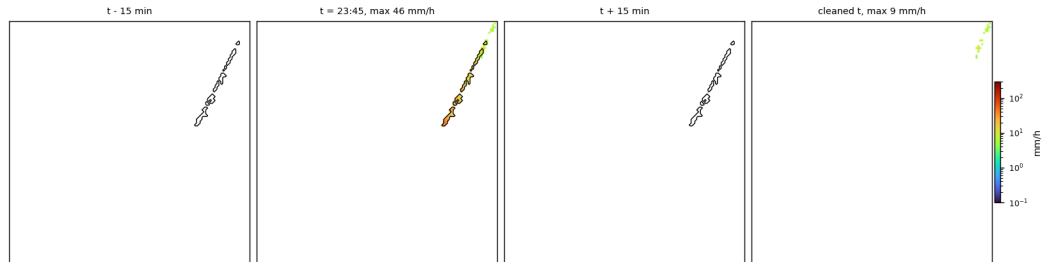
Fails: a radio emitter interfering with a radar draws a thin line pointing at the radar; here inside the Elvira–Friederike storms, 30 May 2016 (black). **Rule:** a thin component ≥ 80 km long, aligned within 10° with the direction to a radar within 250 km, is lowered to its outer-ring median. The convective cells beside it are untouched (max $112 \rightarrow 56 \text{ mm h}^{-1}$).

Example: range ring



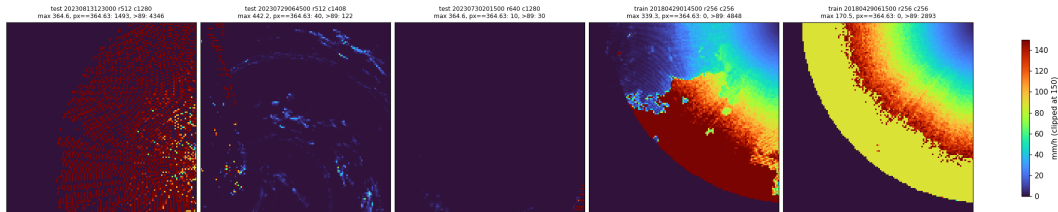
Fails: rings are invisible in most single frames but show up as circles in the yearly frequency of $\geq 31 \text{ mm h}^{-1}$ around a radar (left, 2014; blue: the ring mask). **Rule:** pixels $\geq 89 \text{ mm h}^{-1}$ on the mask are lowered to their outer-ring median ($365 \rightarrow 6 \text{ mm h}^{-1}$). Below 89 mm h^{-1} the gauges see real rain under the rings, so those values stay.

Example: no temporal support



Fails: a 46 mm h^{-1} line appears in one frame only: no echo $\geq 1 \text{ mm h}^{-1}$ within 30 km at $t - 15$ or $t + 15$ min (outlined in all three frames). **Rule:** a cell $\geq 10 \text{ mm h}^{-1}$ without such support is lowered to its outer-ring median. 30 km in 15 min is 33 m s^{-1} , faster than any cell moves. **Gauges:** about 1% of such pixels have rain underneath, in every intensity bin.

Failures that small, local or transient rules cannot see



- ▶ **Reflectivity ceiling at $364.63 \text{ mm h}^{-1} = 64.0 \text{ dBZ}$ ($Z = 200R^{1.6}$):** speckle at one exact value inside real rain, Serbian and neighbouring radars, 2023.
- ▶ **Radar-disk failure, Madrid (Torrejón), 2018-04-29:** a smooth range-only field to 340 mm h^{-1} over the whole disk; 16% of all training pixels above 89 mm h^{-1} .
- ▶ **Ceiling at $48.62 \text{ mm h}^{-1} = 50.0 \text{ dBZ}$, Oradea area, November 2019.**

All three were found by the tail audit: $\xi(u)$ drifted instead of reaching a plateau.

Radar-level checks

Ceiling table (per radar area and year). The count of a value is compared with the largest count of any other value within ± 4 dB; a ceiling stands $\geq 5\times$ above. Calibration: the known ceilings score 9.9–89, all other values have 99.9th percentile 1.96. Ceiling pixels are censored: ≥ 5 in a tile-frame reject it, 1–4 are repaired.

Repeated value (per tile, after repair). Real rain on the 0.01 mm h^{-1} grid almost never repeats a value $\geq 31 \text{ mm h}^{-1}$: per tile, median 1 repeat, 99.9th percentile 24. ≥ 50 rejects the tile; this also catches plateaus left by a repair.

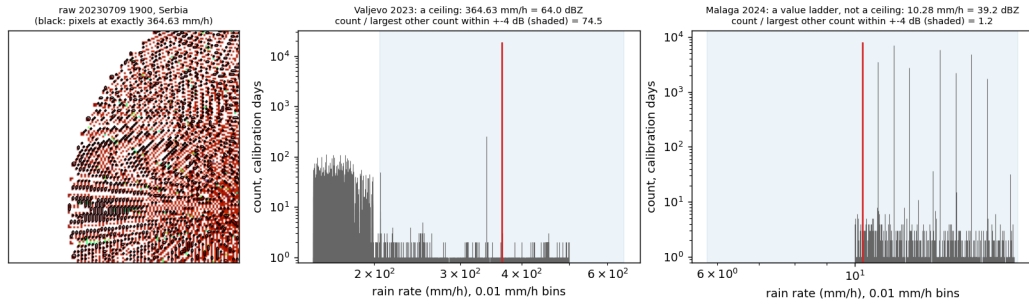
Radar-disk failure (per radar and frame, over the area the radar owns): share of pixels $\geq 31 \text{ mm h}^{-1}$, share of log-rate variance explained by range alone, and the jump in rate across the area boundary and the maximum-range circle. Real rain varies in azimuth.

Radar-day ranking. Each radar-day is scored against the radar's own seasonal 99th percentile and the median of its 5 nearest radars on the same day:

$$\text{score} = \log_{10} \frac{x + f}{\max(q_{99}^{\text{own}}, \text{median}_{\text{neighbours}}) + f}$$

on four signals: share $\geq 150 \text{ mm h}^{-1}$, ceiling pixels, flagged frames, isolated high pixels. The top of the list is looked at; exclusions are decided by eye.

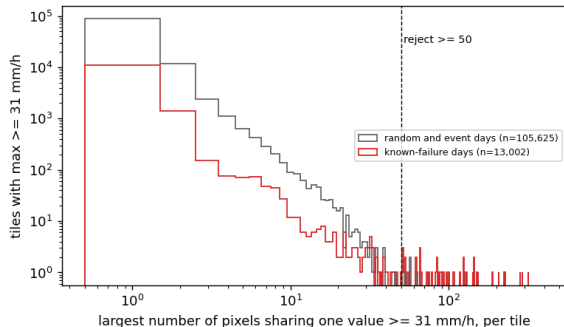
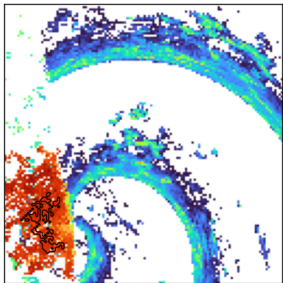
Example: reflectivity ceiling



Fails: speckle at exactly 364.63 mm h^{-1} (64.0 dBZ) inside rain, Serbia 2023 (black). **Rule:** per radar area and year, a value counted $\geq 5\times$ more often than any other value within ± 4 dB. Valjevo: $74.5\times$. Right: a radar that reports on a coarse ladder; each level is about as frequent as the next ($1.2\times$), so it is not a ceiling. Comparing with the median of the neighbouring values instead flagged it at $4,000\times$.

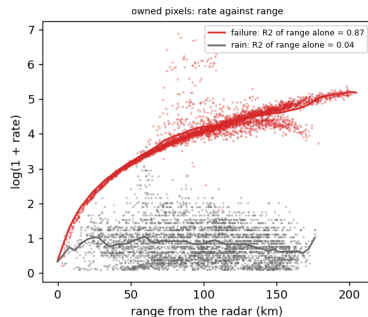
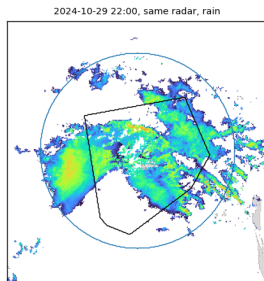
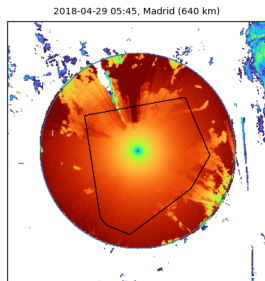
Example: repeated value

20230803 1915: 125 pixels at exactly 107.71 mm/h



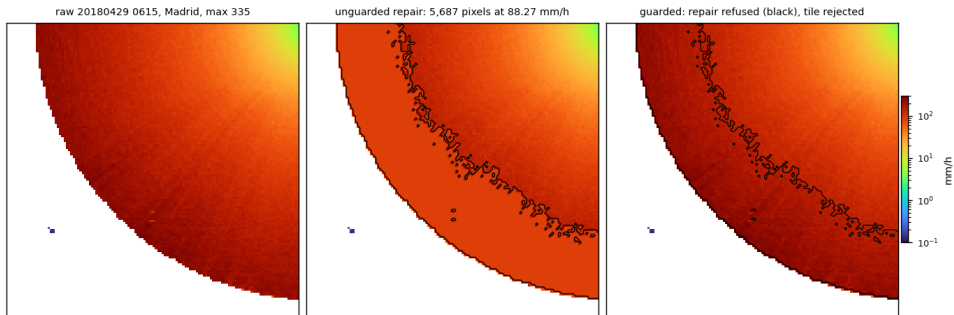
Fails: 125 pixels at exactly 107.71 mm h^{-1} inside a saturated sector (black). **Rule:** on the 0.01 mm h^{-1} grid real rain rarely repeats a value $\geq 31 \text{ mm h}^{-1}$; on random and event days the largest repeat count per tail tile is almost always below 25. A tile with ≥ 50 pixels at one value is rejected. The test also runs after every repair, to catch plateaus a repair creates.

Example: radar-disk failure



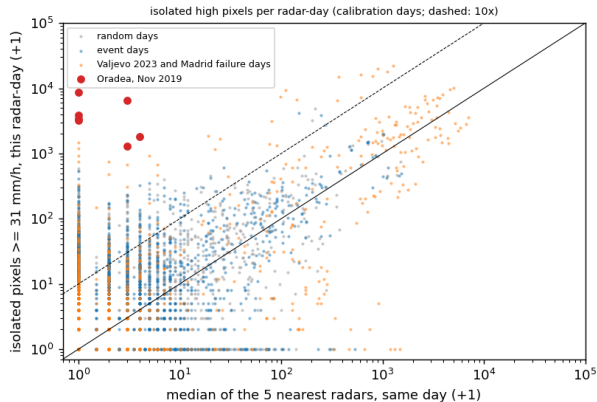
Fails: Madrid, 29 April 2018: the whole disk at a smooth, range-dependent rate up to 340 mm h^{-1} , with a sharp edge at maximum range (blue). Black: the area this radar owns. **Rule:** per radar and frame, the share of log-rate variance explained by range alone (failure 0.87, rain 0.04), the share $\geq 31 \text{ mm h}^{-1}$, and the jump at the area boundary and the maximum-range circle. A flagged radar-frame rejects every tile it owns $\geq 10\%$ of.

Example: guarded repair



Fails: the footprint repair lowers a region to the median of its outer ring. When the ring belongs to the same failing radar, the repair produces a flat plateau (middle: 5,687 pixels at 88.27 mm h^{-1}) and hides the failure. **Rule:** a component above 500 pixels is not repaired (black) and the tile is rejected; the repeated-value test after every repair is the second safeguard.

Example: radar-day ranking



Fails: Oradea, November 2019: clusters of 2–9 pixels at $40\text{--}340\text{ mm h}^{-1}$ in dry surroundings, which the spike rule cannot see. **Rule:** isolated pixels $\geq 31\text{ mm h}^{-1}$ per radar-day against the 5 nearest radars on the same day. Oradea (red): 1,286–8,636 against 0–3. Event radar-days whose neighbours have almost none stay below a few hundred. The top of the ranking is looked at before anything is excluded.

Splits

- ▶ **Blocked in time:** whole ISO weeks drawn per calendar month, 1-day buffers; every test day is at least 2 days from any training day.
- ▶ **Documented extremes:** 27 storms from published case studies (MeteoSwiss, DWD, ESSL, AEMET, Météo-France, ECMWF), with their whole week, kept out of training (25 in test, 2 in the NIMBUS split); every tile kept.
- ▶ **Product shift:** train, validation and test are ODYSSEY only; NIMBUS days are a separate test split.
- ▶ Stratified sampling by tile maximum with inclusion weights, so the tail is oversampled and statistics can be reweighted.

	train	validation	test	NIMBUS
patches	1,962,057	252,809	297,460	487,673
event tiles	–	–	36,166 (25 events)	5,249 (2 events)

Sizes of the current build (before the radar-level checks).

Validation against gauges: design

- ▶ 1,735 DWD and SwissMetNet 10-min gauges, 12.1M station-frames on 4,968 days. The radar frame matches the gauge interval 10–20 min later (log-rate correlation 0.29–0.47).
- ▶ Gauge quality control on the gauges alone: fault codes, stuck runs, isolated bursts with no rain at any gauge within 30 km.
- ▶ **Presence**: does the gauge see rain ($\geq 1 \text{ mm h}^{-1}$) under a pixel? Per rule and intensity bin.
- ▶ **Values**: exceedance ratio over the same station-frames, with no conditioning on either side:

$$\rho(u) = \frac{N(\text{radar} \geq u)}{N(\text{gauge} \geq u)}$$

A flat ρ means the radar has the gauges' tail *shape*. Its level mixes point-vs-area and instantaneous-vs-10-min differences, which are not quantified.

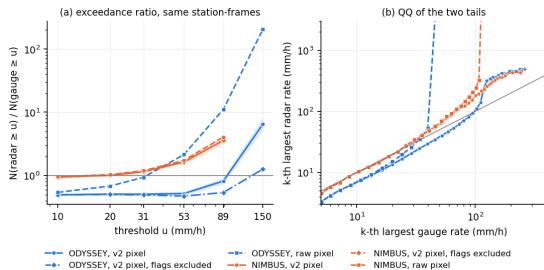
Presence: the rules remove artefacts, tile rejection removes rain

Share of pixels with gauge $\geq 1 \text{ mm h}^{-1}$, by radar intensity:

class	[10,31)	[31,89)	[89,150)	[150,500)	≥ 500
untouched	0.91	0.92	0.91	0.90	–
no temporal support	0.01	0.01	0.00	0.00	0.01
range ring	0.82	0.50	0.00	0.00	0.00
spike repaired	0.13	0.24	0.24	0.26	0.00
other pixels of tiles rejected as unphysical	0.81	0.74	0.55	0.10	0.05

- ▶ Temporal support is the cleanest rule. Rings carry real rain below 31 mm h^{-1} and none from 89 mm h^{-1} , hence ring repair only from 89 mm h^{-1} .
- ▶ A tile rejected for one bad pixel is usually a real storm: repair the pixel instead.
- ▶ Repaired pixels: the gauge saw $\geq 10 \text{ mm h}^{-1}$ under 5.3% of them (untouched: 44.6%).
- ▶ The quality index does not separate real from false tail pixels.

Values: is the tail calibrated?



$\rho(u)$	31	89	150
ODYSSEY raw	0.93	11.0	204
ODYSSEY, cleaned	0.51	0.82	6.5
+ flags excluded	0.49	0.54	1.26
+ pixel repairs	–	0.57	1.47
NIMBUS, cleaned	1.14	3.6	31

- ▶ **ODYSSEY:** the radar exceeds u about half as often as the gauges, at every level up to 89 mm h^{-1} once flagged pixels are excluded. The tail shape matches; an excess of $2.5\text{--}3\times$ remains at 150 mm h^{-1} (≈ 100 gauge exceedances).
- ▶ **NIMBUS:** heavier than the gauges from $\approx 31 \text{ mm h}^{-1}$. The pixels are rain (96% have rain at the gauge at $150\text{--}500 \text{ mm h}^{-1}$), and no rule touches them.
- ▶ At gauge $\geq 30 \text{ mm h}^{-1}$, median radar/gauge ratio 0.45.

ODYSSEY and NIMBUS on the same days

101 days held in both products (2,407 frames, 97,682 wet tiles):

quantity	NIMBUS / ODYSSEY
wet area $\geq 0.1 / 1 \text{ mm h}^{-1}$	0.71 / 0.99
exceedance $\geq 10 / 31 / 89 \text{ mm h}^{-1}$	1.36 / 1.44 / 2.4
perimeter of the $10\text{--}31 \text{ mm h}^{-1}$ excursion sets	≈ 1.8
spectral power at 128 / 32 / 8 / 4 km	1.25 / 1.42 / 3.1 / 6.2
+15 min frame correlation	0.41 $\rightarrow$ 0.26

NIMBUS fields are more intense, rougher and more fragmented at small scales, consistent with an instantaneous scan against a 15-min composite. A model trained on ODYSSEY pays a product penalty on NIMBUS that is unrelated to extremeness, so NIMBUS is scored against same-product references only.

When is the dataset ready? A gate fixed in advance

The dataset is ready when one more cleaning step would not change any reported number.

1. **Known failures absent**, from a regression catalogue.
2. **Residual contamination measured**: 100 random test tiles with $\max \geq 89 \text{ mm h}^{-1}$ looked at; 95% upper bound on the artefact rate $\leq 5\%$.
3. **Stable reference**: dropping the 20 most influential non-event days moves no observed tail statistic outside its day-block bootstrap interval.
4. **Real extremes survive**: $< 0.5\%$ of tiles rejected per event; no rule's rejection rate rises steeply with intensity; gauge ρ no worse than before.
5. **Remaining drift in $\xi(u)$ explained** by a named cause (truncation, regional mixing, sub-asymptotic threshold).

Calibration so far (199 days: known failures, 27 events, 100 random days): every known failure caught (the remaining tiles of those days checked by eye); worst event 0.14% of tiles rejected; random-day rejection $\leq 0.014\%$ below 150 mm h^{-1} . The full-archive radar pass is running.

Open issues

- ▶ **Isolated high clusters.** Clusters of 2–9 pixels at $40\text{--}340\text{ mm h}^{-1}$ in dry surroundings survive on the Oradea days. A per-tile rule would also remove 9–16% of event tiles, so they are handled per radar-day, against the radar's neighbours.
- ▶ **$486.2\text{ mm h}^{-1} = 66.0\text{ dBZ}$** is the maximum of 55 tiles; no nearby value is the maximum of more than 3. For data in 0.5 dBZ steps it is the highest level below our 500 mm h^{-1} bound ($66.5\text{ dBZ} = 522.6\text{ mm h}^{-1}$ is repaired), so the bound censors these radars at 66 dBZ. At least one case sits on an artefact arc.
- ▶ **$11.53\text{ mm h}^{-1} = 40.0\text{ dBZ}$** piles up on three of the radars that carry the 64 dBZ ceiling in 2023.
- ▶ **NIMBUS tail excess** over the gauges, untouched by any rule.
- ▶ **Quality index inverted under ODYSSEY:** unconfirmed tail pixels have the higher index (median 0.90 vs 0.20).
- ▶ **Validation covers Germany and Switzerland only;** the failures above were in Serbia, Spain and Romania.

What the dataset is for: the hypothesis

Minkowski functionals (area, perimeter, Euler characteristic of the excursion sets $\{x \geq u\}$ across thresholds) describe the geometry of the rain field. If an extreme storm has the geometry of an ordinary one at higher intensity, a model trained on these functionals should carry that geometry beyond its training range.

	claim	tested by
H1	excursion-set geometry of extremes, suitably normalised, resembles that of ordinary events	data analysis, no training
H2	the structural loss ties geometry at high thresholds to geometry learned at lower ones	gradient and residual analysis
H3	beyond the training range, a model trained with it produces more realistic extreme structure than one trained without	cut-off experiment

So far: on the old data the loss helps the deterministic model's tail substantially (exceedance ratio 0.12 $\rightarrow$ 0.24) for +3.8% MAE. Those results used splits that leak between train and test, so they are interpolation scores and must be redone on the new data.

Next steps, once the dataset passes its gate

	step	what it settles	needs
1	Retrain the baselines	deterministic models with and without the Minkowski loss; flow matching on the best one, with the loss as a reward	dataset
2	Fix the extreme threshold	the peaks-over-threshold level on screened data: stability of $\xi(u)$, the bound, regional and seasonal levels	dataset
3	Fair loss comparison	spectral, SSIM, wet-area, optical flow vs Minkowski at matched gradient share, same budget, last checkpoint, 3 seeds	1
4	H1: is extreme geometry ordinary?	do normalised functional curves of extremes collapse onto those of ordinary storms, by region and season?	dataset
5	H3: cut-off experiment	train without every space-time block above a cut, test on the held-out extremes; i.i.d. to o.o.d. degradation with and without the loss; area only vs all three functionals	1, 2, 4
6	Regime-aware loss	should regimes, recognised from the coarse input, set the loss's thresholds or weights?	4
7	Robustness	NIMBUS product shift; possibly Switzerland held out, or NWP input	5
8	Release	a Europe-wide benchmark: splits, event and product-shift test sets, quality flags, code to regenerate	1, 2

Questions: artefacts and screening

1. Which artefact types does MeteoSwiss see in the OPERA composite that this screen misses? Which composite-level filters do you trust (texture, satellite cloud masks, others), and with which thresholds?
2. Are **reflectivity ceilings** at round dBZ values (64.0, 50.0, 40.0 dBZ) known for OPERA? Are they receiver saturation, a processing limit, or something else?
3. How do you tell a **small intense core** from a **cluster of clutter pixels** without volume data? Is a radar-day comparison with neighbouring radars a sensible test?
4. **Repair or reject?** For ML training sets, do you repair artefact pixels or drop the sample? Any experience of models learning repaired values?
5. Why would the OPERA **quality index** be higher on unconfirmed tail pixels under ODYSSEY, and carry no signal under NIMBUS? Is it usable in any form?

Questions: tail calibration and plausibility

6. Is a factor ≈ 0.5 between ODYSSEY and gauge exceedance counts expected, from Z-R and from comparing a 2 km pixel with a point?
7. **NIMBUS rates heavy rain above the gauges**, increasingly with intensity ($1.15\times$ the gauge exceedances at 31 mm h^{-1} , $3.6\times$ at 89 mm h^{-1}). Is this known? Instantaneous scan vs 10-min totals, hail, or the NIMBUS processing? This decides whether NIMBUS can serve as a test set.
8. What is a defensible **upper plausibility bound** for a 15-min, 2 km rate? We use 500 mm h^{-1} , with repair down to 150 mm h^{-1} around such cores. On data in 0.5 dBZ steps it acts at 66.0 dBZ (486 mm h^{-1}).
9. Should a **hail cap** (e.g. above $\approx 55 \text{ dBZ}$) be applied before any tail statistic?
10. How should the **product breaks** (2015, 2017, 2024) be handled when pooling years for training?

Questions: defining and measuring extremes

11. How does MeteoSwiss define intense precipitation for **radar rates** at 2 km and 5–15 min, as opposed to accumulation-based warning levels? Is a fixed 31 mm h^{-1} meaningful, or should the level be regional, seasonal or product-specific?
12. **Rate or accumulation?** Should the tail be defined on 1-h accumulations, which are impact-relevant and comparable with gauges? 15-min snapshots miss short peaks (Valencia 2024: 28.6 mm h^{-1} at the radar pixel vs a 184.6 mm h^{-1} gauge-hour).
13. **Peaks over threshold from radar:** how do you choose the threshold, decluster exceedances in space and time, and choose between pixel-level and event-maximum fits? How do you report return levels from radar?

Questions: independent truth and out-of-distribution tests

14. Can we access **CombiPrecip and POH/MESHS for 2012–2026**? The open portal keeps 14 days.
15. Which **shift** matters operationally for downscaling: unprecedented intensities, Alpine orography, new radars or products, future climate, or NWP input (ICON-CH1/2)?
16. Would **Switzerland held out** be a meaningful test, with CombiPrecip and the gauges as independent truth? Is a held-out region meaningful when every country has its own network?
17. Above which rate does the radar tail stop being trustworthy as a test target for an **intensity cut-off** experiment? In mm/h or relative to local climate?
18. Is there an **event catalogue** (TRT tracks, hail, lightning) that could label storm types across the archive? How does MeteoSwiss evaluate its own ML nowcasting on unseen extremes?
19. Would MeteoSwiss **validate the Swiss part** of a released benchmark, or co-own it?

The three answers that unblock the most

1. **The plausibility bound and the hail cap** (Q8, Q9): with 0.5 dBZ steps the bound censors the tail at 486 mm h^{-1} ; together they decide where the tail ends and how it is fitted.
2. **The NIMBUS excess** (Q7): it decides whether the NIMBUS split is a product robustness test or a biased reference.
3. **CombiPrecip and POH/MESHS access** (Q14): an independent truth over complex terrain, for validating the screen and for an out-of-distribution test.